Semester project

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RQ:

Does a high carbon tax rate lead to lower GHG emissions per capita after factoring in fossil fuel share in the energy mix and GDP?

packages installed:

install.packages("tidyverse") install.packages("tinytex") install.packages("hrbrthemes") install.packages("ggrepel")

```
library(tidyverse)
library(readxl)
library(ggplot2)
library(hrbrthemes)
library(ggrepel)


Carbon_tax <- read_excel("Carbon_tax.xlsx", sheet = "Compliance_Price", skip = 1)
GHG <- read_csv("GHG.csv")
Population <- read_csv("Population.csv")
GDP <- read_csv("GDP.csv")
Energy_mix <- read_csv("Energy_mix.csv")
```

(Tax rate is USD per tonne of CO2-equivalent)

```
#Choosing and renaming relevant columns
My_countries <- c("Austria", "Belgium", "Denmark", "Estonia", "Finland",
                  "France", "Germany", "Greece", "Hungary", "Iceland",
                  "Ireland", "Italy", "Latvia", "Lithuania", "Luxembourg",
                  "Netherlands", "Norway", "Poland", "Portugal",
```

```
                  "Slovenia", "Spain", "Sweden")

Year_range <- 2005:2024

Carbon_tax_clean <- Carbon_tax %>%
  filter(
    `Instrument Type` == "Carbon tax"
    ) %>% #filtering out ETS only keeping carbon tax
  mutate(
    Country = str_extract(`Name of the initiative`,
  paste(My_countries, collapse = "|"))
  ) %>%
  filter(!is.na(Country)) %>%
  select(
    Country, all_of(as.character(Year_range))
    ) %>%
  pivot_longer(
    cols = -Country, names_to = "Year", values_to = "Tax_rate_usd"
    ) %>%
  mutate(
    Year = as.numeric(Year),
    Tax_rate_usd = ifelse(is.na(Tax_rate_usd), 0, Tax_rate_usd)
    )

no_tax_countries <- setdiff(My_countries, unique(Carbon_tax_clean$Country))

no_tax_filled <- expand_grid(Country = no_tax_countries, Year = Year_range
    ) %>%
  mutate(Tax_rate_usd = 0)

Carbon_tax_clean <- bind_rows(Carbon_tax_clean, no_tax_filled) %>%
  arrange(Country, Year)

#Choosing and renaming relevant columns
GHG_clean <- GHG %>%
  select(
    `Geopolitical entity (reporting)`, TIME_PERIOD, OBS_VALUE
    ) %>%
  rename(
    Country = `Geopolitical entity (reporting)`,
    Year = TIME_PERIOD,
    Total_emissions = OBS_VALUE
```

```
  )

#Choosing and renaming relevant columns
Population_clean <- Population %>%
  select(
    `Geopolitical entity (reporting)`, TIME_PERIOD, OBS_VALUE
  ) %>%
  rename(Country = `Geopolitical entity (reporting)`,
         Year = TIME_PERIOD,
         Population = OBS_VALUE
         )
#Choosing and renaming relevant columns
GDP_clean <- GDP %>%
  select(
    `Geopolitical entity (reporting)`, TIME_PERIOD, OBS_VALUE
  ) %>%
  rename(
    Country = `Geopolitical entity (reporting)`,
    Year = TIME_PERIOD,
    GDP_per_capita = OBS_VALUE
  )

#Choosing and renaming relevant columns
Energy_mix_clean <- Energy_mix %>%
  select(
    Country = `Geopolitical entity (reporting)`,
    Year = TIME_PERIOD,
    Source = `Standard international energy product classification (SIEC)`,
    OBS_VALUE
  ) %>%
  filter(Source %in% c("Solid fossil fuels",
  "Natural gas",
  "Oil and petroleum products (excluding biofuel portion)", "Total")
         ) %>%
  pivot_wider(
    names_from = Source, values_from = OBS_VALUE, values_fill = 0
    ) %>%
  mutate(
    Fossil_fuels_share = (`Solid fossil fuels` + `Natural gas` +
    `Oil and petroleum products (excluding biofuel portion)`)
    / Total * 100
    ) %>%
```

```
  select(
    Country, Year, Fossil_fuels_share
    )
```

```
Merged1 <- merge(
  Carbon_tax_clean, GHG_clean,
  by.x = c("Country","Year"), by.y = c("Country", "Year")
  )

Merged2 <- merge(
  Population_clean, GDP_clean,
  by.x = c("Country","Year"), by.y = c("Country", "Year")
  )

Merged3 <- merge(
  Merged1, Merged2,
  by.x = c("Country","Year"), by.y = c("Country", "Year")
  )

Trial_dataset <- merge(
  Merged3, Energy_mix_clean,
  by.x = c("Country","Year"), by.y = c("Country", "Year")
  )
```

Calculating: emissions per capita = Total emissions/ Total population (Emissions per capita is measured in tonnes of CO2-equivalent per person)

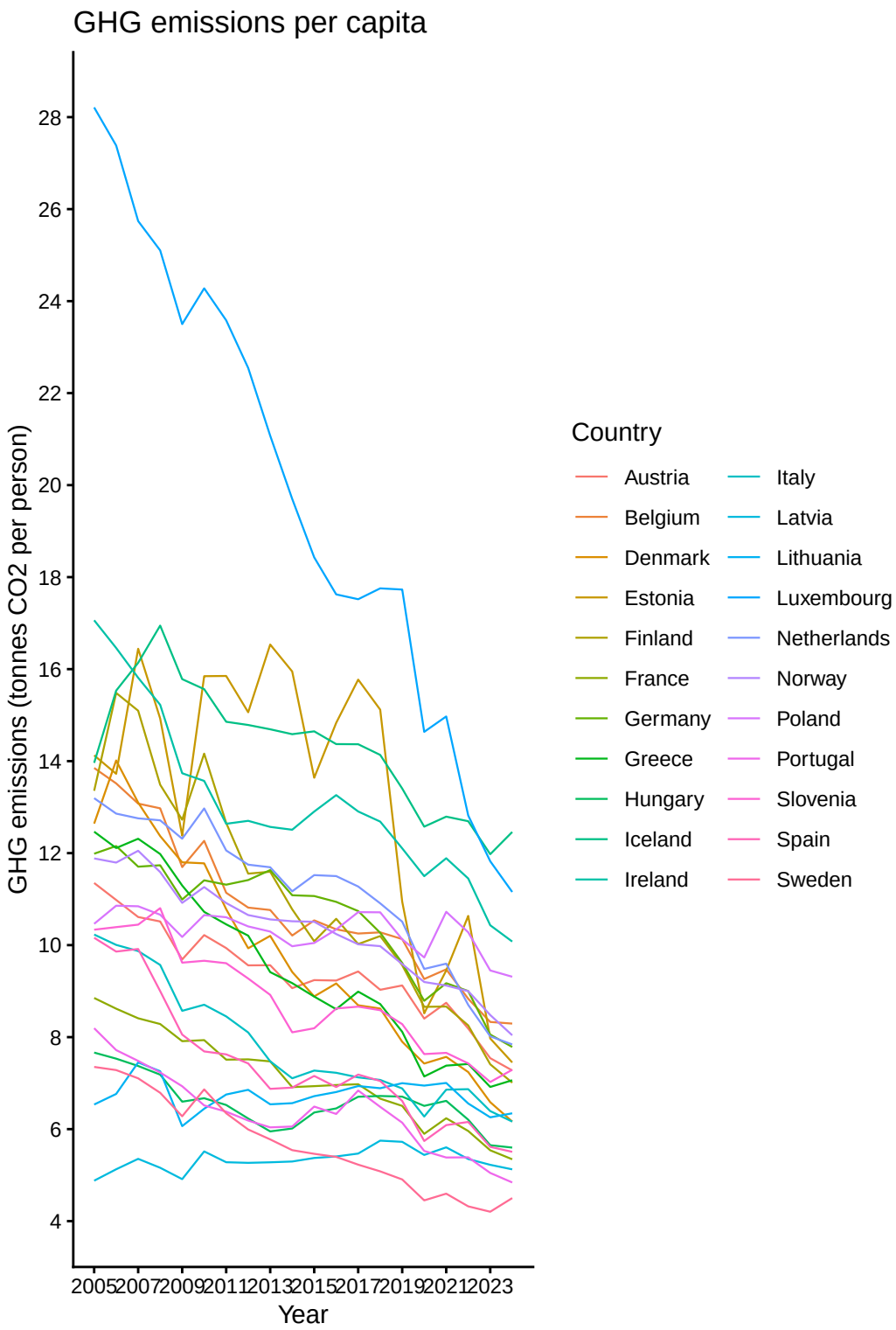
Unit of measure is in tonnes

```
Trial_dataset$Emissions_per_capita <-
  Trial_dataset$Total_emissions * 1000000 / Trial_dataset$Population

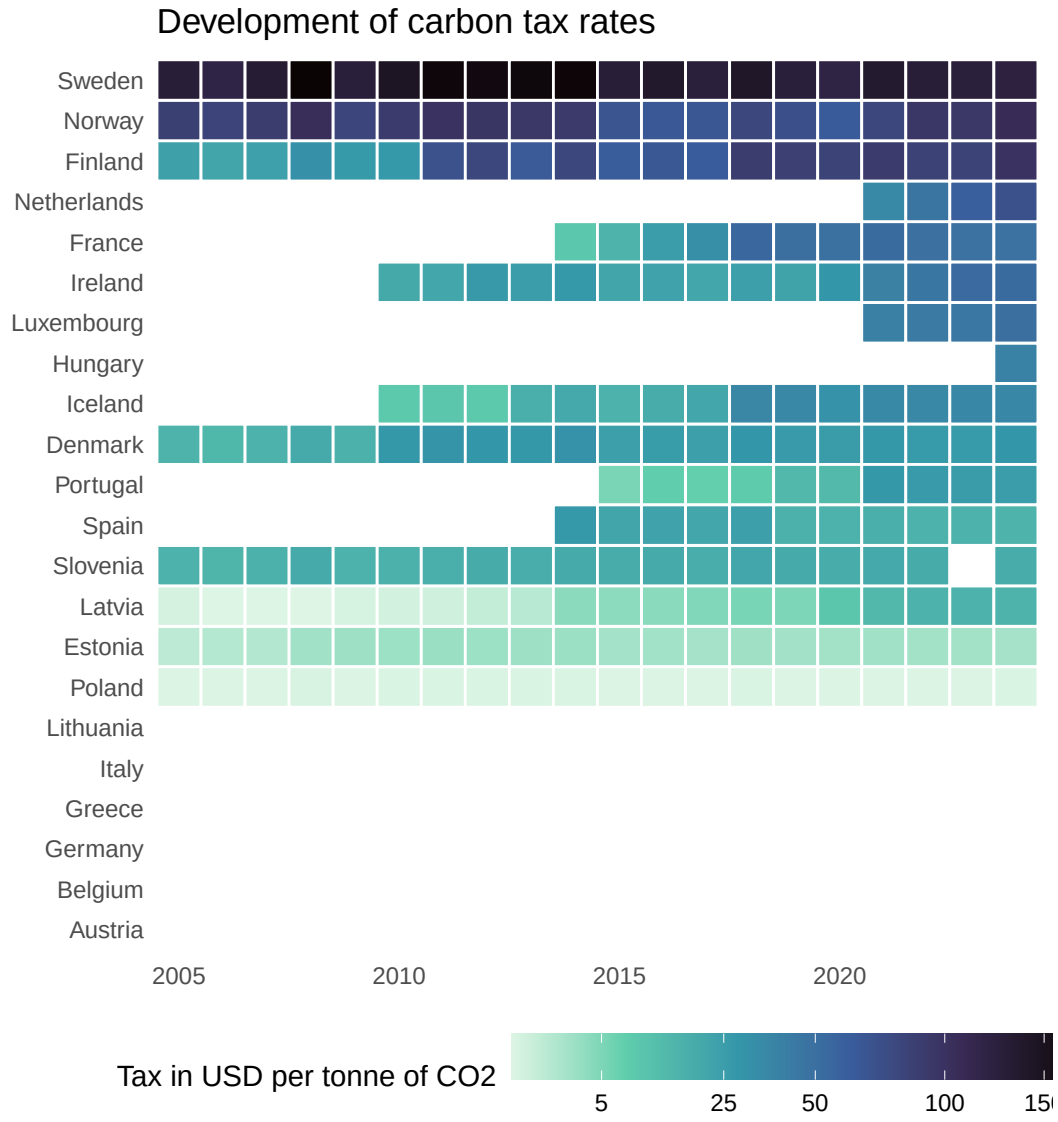
Final_dataset <- Trial_dataset %>%
  select(
    Country, Year, Tax_rate_usd, GDP_per_capita, Fossil_fuels_share,
    Emissions_per_capita
  )
```

Plotting each countrie GHG emissions per capita throughout the years. How did emissions evolve throughout the years?

```r
ggplot(
  Final_dataset, aes(x = Year, y = Emissions_per_capita, color = Country)) +
  geom_line(linewidth = 0.4) +
  scale_x_continuous(breaks = seq(2005, 2024, by = 2)) +
  scale_y_continuous(breaks = seq(0, 30, by = 2)) +
  labs(
    title = "GHG emissions per capita", y = "GHG emissions (tonnes CO2 per person)"
  ) +
  theme_classic(base_size = 11)
```


GHG emissions per capita
GHG emissions (tonnes CO2 per person)
28
26
24
22
20
18
16
14
12
10
8
6
4
2005
2007
2009
2011
2013
2015
2017
2019
2021
2023
Year
Country
Austria
Belgium
Denmark
Estonia
Finland
France
Germany
Greece
Hungary
Iceland
Ireland
Italy
Latvia
Lithuania
Luxembourg
Netherlands
Norway
Poland
Portugal
Slovenia
Spain
Sweden

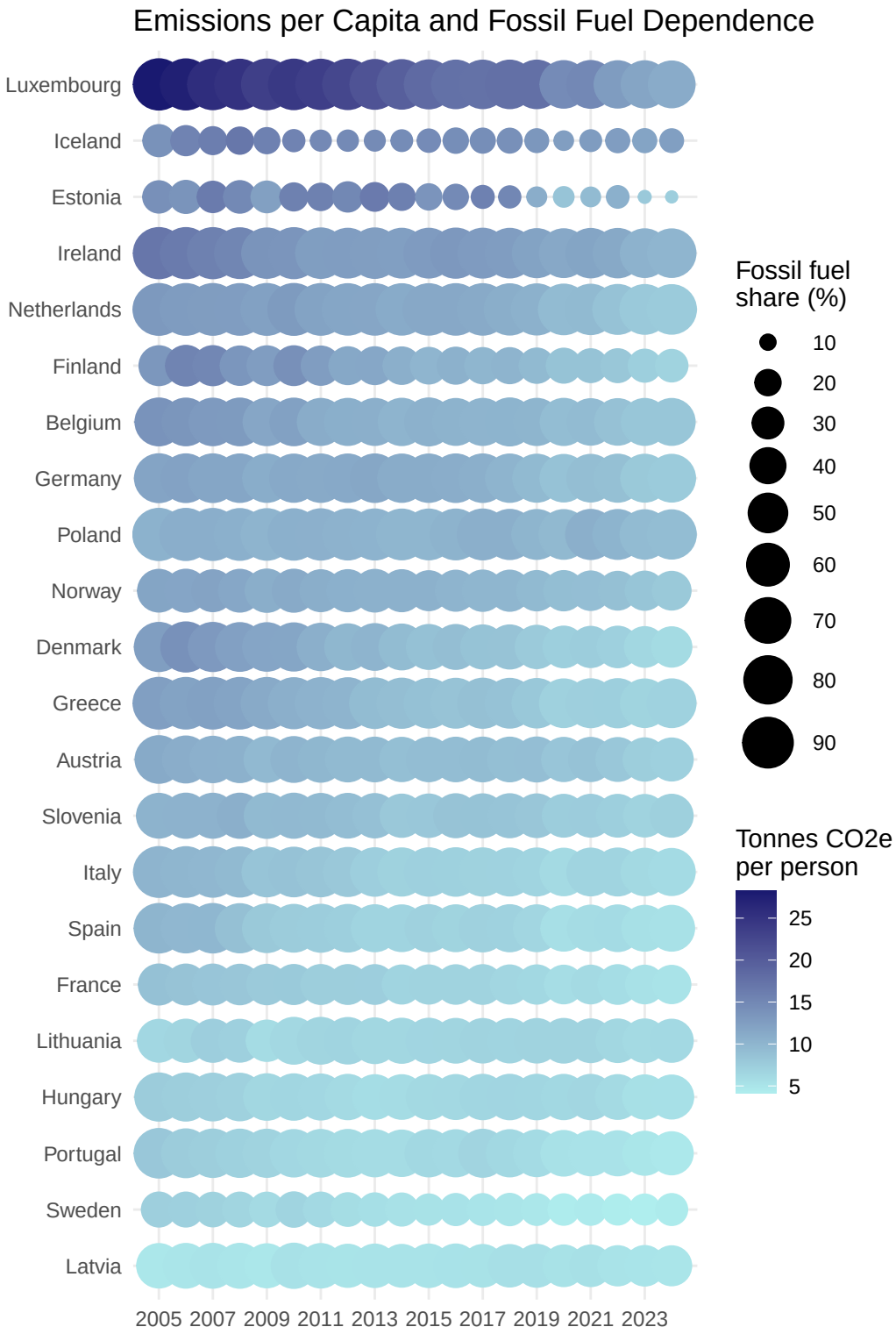
```r
Final_dataset %>%
   mutate(
    Carbon_Tax = ifelse(Tax_rate_usd <= 0, NA, Tax_rate_usd),
    Country = fct_reorder(Country, Tax_rate_usd, .fun = max)
  ) %>%
  ggplot(aes(Year, Country, fill = Carbon_Tax)) +
  geom_tile(colour = "white", linewidth = 0.5) +
   theme_minimal(base_size = 11) +
  scale_x_continuous(breaks = seq(2005, 2024, by = 5), expand = c(0, 0)) +
  labs(x = NULL, y = NULL) +
    theme(panel.grid = element_blank(), legend.position = "bottom", legend.key.width = unit(
  ) +
  scale_fill_viridis_c(
    option = "mako", direction = -1, trans = "sqrt",
    breaks = c(5, 25, 50, 100, 150),
    na.value = "white",
    name = "Tax in USD per tonne of CO2"
  ) +
  labs(title = "Development of carbon tax rates")
```



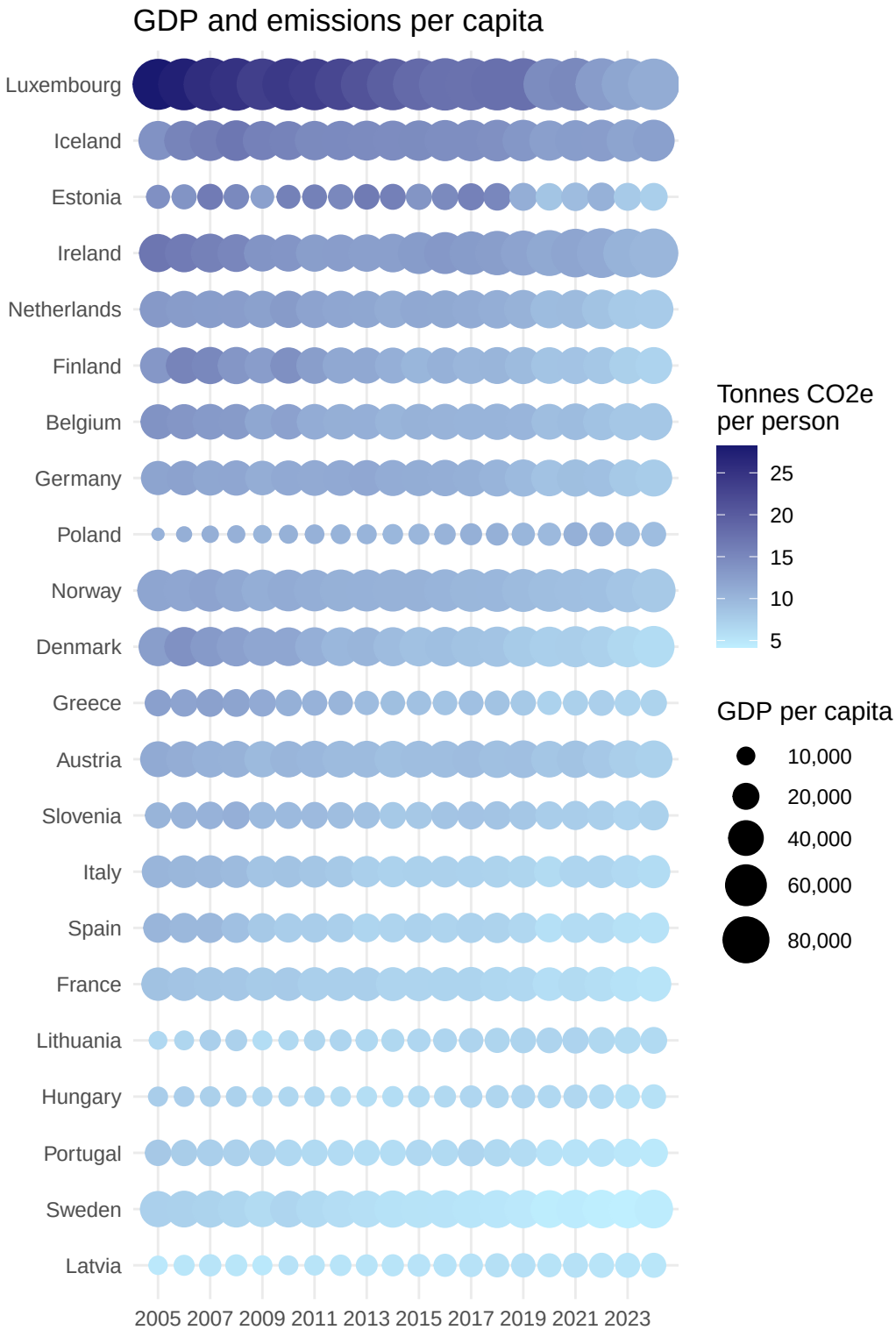
#PLotting fossil fuels share in the countries energy mix, and in a seperte plot emissions agains GDP.

```
Final_dataset %>% #Putting the countries in order from highest to lowest e, and getting the a
  mutate(Country = reorder(Country, Emissions_per_capita, FUN = mean)) %>%
  ggplot(aes(Year, Country, size = Fossil_fuels_share, colour = Emissions_per_capita)) +
  geom_point(alpha = 1) +
  scale_colour_gradient(low = "paleturquoise", high = "midnightblue", breaks = seq(0, 30, by
  scale_size(range = c(2, 10), breaks = seq(0, 100, by = 10),
             name = "Fossil fuel\nshare (%)") +
```

```
  scale_x_continuous(breaks = seq(2005, 2024, by = 2)) +
  labs(x = NULL, y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(), legend.position = "right") +
  labs(title = "Emissions per Capita and Fossil Fuel Dependence")
```


Emissions per Capita and Fossil Fuel Dependence
Luxembourg
Iceland
Estonia
Ireland
Netherlands
Finland
Belgium
Germany
Poland
Norway
Denmark
Greece
Austria
Slovenia
Italy
Spain
France
Lithuania
Hungary
Portugal
Sweden
Latvia
2005 2007 2009 2011 2013 2015 2017 2019 2021 2023
Fossil fuel share (%)
10
20
30
40
50
60
70
80
90
Tonnes CO2e per person
25
20
15
10
5

```r
#Order of countries by average emissions (lowest at bottom, highest at top)
Final_dataset %>%
  mutate(Country = reorder(Country, Emissions_per_capita, FUN = mean)) %>%
  ggplot(aes(Year, Country, size = GDP_per_capita, colour = Emissions_per_capita)) +
  geom_point(alpha = 5) +
  scale_colour_gradient(low = "lightblue1", high = "midnightblue", breaks = seq(0, 30, by = 5
  scale_size(range = c(2, 10), breaks = c(10000, 20000, 40000, 60000, 80000, 110000), labels
             name = "GDP per capita") +
  scale_x_continuous(breaks = seq(2005, 2024, by = 2)) +
  labs(x = NULL, y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(), legend.position = "right") +
  labs(title = "GDP and emissions per capita")
```


GDP and emissions per capita
Luxembourg
Iceland
Estonia
Ireland
Netherlands
Finland
Belgium
Germany
Poland
Norway
Denmark
Greece
Austria
Slovenia
Italy
Spain
France
Lithuania
Hungary
Portugal
Sweden
Latvia
2005 2007 2009 2011 2013 2015 2017 2019 2021 2023
Tonnes CO2e per person
25
20
15
10
5
GDP per capita
10,000
20,000
40,000
60,000
80,000